Multiple Choice Quiz

Internet-Scale Naming Architecture

Total Questions: 10
Time Allowed: 20 minutes
Instructions: Choose the **best** answer for each question. Each question carries equal marks.

Question 1

Why is a single global directory technically unfeasible for internet-scale naming?

A. It would require too much human administrative effort to maintain
B. It would make name resolution too fast for distant users
C. Centralization improves performance for all users equally
D. It causes high geographical latency for distant users and lacks sufficient memory/processing capacity at billion-user scale

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Question 2

What characterizes the "Global Layer" in the three-layer partitioning of a name space?

A. It changes frequently and requires immediate updates
B. It is highly stable, rarely changes, supports effective caching, and uses "lazy" updates
C. It operates only at the departmental level with no replication
D. It focuses exclusively on fast, real-time responsiveness

Question 3

Explain the primary difference between logical layers and physical zones.

A. Logical layers categorize data by stability and scale conceptually, while physical zones represent actual infrastructure and non-overlapping parts of the namespace managed by specific servers
B. Logical layers and physical zones are identical concepts
C. Physical zones are conceptual, while logical layers describe real servers
D. Logical layers only apply to DNS, while physical zones apply to LDAP

Question 4

How does the core question asked by Structured Naming (DNS) differ from that of Attribute-Based Naming?

A. Both ask the same fundamental question about entity location
B. Attribute-Based Naming asks “Where is it?”, while Structured Naming asks “What is it?”
C. Structured Naming asks “Where is it?” using unique paths, while Attribute-Based Naming asks “What is it?” using attribute-value pairs
D. Structured Naming is used only for email, while Attribute-Based Naming is used for files

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Question 5

What is the “Search Cost” associated with hierarchical discovery in systems like LDAP?

A. It allows direct path-based lookups with zero coordination cost
B. It reduces network traffic by using aggressive caching at every level
C. It only affects leaf nodes and never impacts higher-level servers
D. It requires exhaustive top-down searching of every leaf node, causing expensive network coordination and high computational burden

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Question 6

Identify and briefly describe the three primary bottlenecks of decentralized indexing.

A. Communication overload (contacting many servers), client processing drain (filtering millions of records locally), and range query failure (hashing struggles with attribute ranges)
B. Only one bottleneck exists: lack of encryption
C. The main issues are slow hardware and poor user interfaces
D. Bottlenecks only occur in centralized systems, not decentralized ones

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Question 7

What is the specific goal of using space-filling curves to flatten dimensions?

A. To increase the number of dimensions for more complex queries
B. To map N-dimensional attribute space into a 1D linear index while preserving data locality relationships
C. To completely eliminate the need for any indexing
D. To make all data globally visible without any partitioning

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Question 8

How do Hilbert curves maintain data locality when mapping coordinates to a linear path?

A. They randomly scatter points across the line to improve load balancing
B. They use straight vertical and horizontal lines only
C. They recursively divide space into quadrants and connect them with a continuous line so that proximity in multi-dimensional space becomes proximity on the 1D curve
D. They ignore spatial relationships entirely

Question 9

What are the three steps required to map multi-dimensional attributes to storage in a Distributed Hash Table (DHT)?

A. Encrypt the data, replicate it three times, then store it randomly
B. Convert to XML, compress the file, then upload to the cloud
C. Hash the attributes directly without normalization or curves
D. Normalize attribute values to [0, 1), index them on a high-order Hilbert curve to create a 1D index, then map the indices to a DHT identifier space (e.g., Chord ring)

Question 10

Describe the "Dimensionality Cycle" as it pertains to the evolution of naming architectures.

A. It starts with 1D structured paths (DNS), moves to N-dimensional attributes (LDAP) for richer discovery, then mathematically reduces N-dimensions back to 1D using space-filling curves for scalable decentralization
B. It is a linear progression that only moves from simple to complex without returning
C. It refers only to the physical layout of hard drives
D. It describes the cycle of adding and removing users from a system

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Answer Key

Question	Correct Answer	Explanation
1	**D**	A single global directory causes high latency for distant users and cannot handle the memory/traffic volume at internet scale.
2	**B**	The Global Layer is highly stable, supports strong caching, and uses lazy updates.
3	**A**	Logical layers are conceptual (based on stability/scale). Physical zones describe actual server infrastructure.
4	**C**	Structured Naming asks "**Where is it?**" (paths). Attribute-Based Naming asks "**What is it?**" (attributes).
5	**D**	Hierarchical discovery in LDAP requires expensive exhaustive top-down searches across all leaf nodes.
6	**A**	The three bottlenecks are: communication overload, client processing drain, and inability to handle range queries.
7	**B**	Space-filling curves map N-dimensional data to 1D while preserving locality relationships.
8	**C**	Hilbert curves recursively divide

Question	Correct Answer	Explanation
		space and connect quadrants so multi-dimensional proximity becomes linear proximity.
9	**D**	The three steps are: normalize to [0,1), apply Hilbert curve indexing, then map to DHT identifier space.
10	**A**	The Dimensionality Cycle moves from 1D paths → N-D attributes → back to 1D via curves for scalable systems.

End of Quiz

Correct Answer Distribution: A = 3 | B = 2 | C = 2 | D = 3
This quiz covers the three-layer architecture, structured vs. attribute-based naming, bottlenecks, Hilbert curves, and the Dimensionality Cycle.